Status	Finished
Started	Sunday, 9 August 2026, 4:09 PM
Completed	Sunday, 9 August 2026, 4:23 PM
Duration	13 mins 57 secs
Marks	58.68/60.00
Grade	**9.78** out of 10.00 (**97.79**%)

Question 1

Correct

Mark 30.00 out of 30.00

To convert 19.6875 to IEEE 754 floating-point format step by step:

1. Sign bit: 0
2. Convert to binary:
   2.1. Integer 19 → 10011
   2.2. Fraction 0.6875 → 0.6875×2= 1 → 0.375 ×2=0 → 0.75 ×2=1 → 0.5 ×2=1 then the binary number is 1011
   2.3. 19.6875 binary number is 10011 . 1011
3. Normalize: 1.00111011 × 2^ 4
4. Exponent: 4 + 127 = 131 → 10000011
5. Mantissa: 00111011000000000000000
6. IEEE 754 (32-bit): MSB 0 1 0 0 0 0 0 0 1 1 0 0 0 1 1 1 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 LSB

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1 0

10011 1011 11001 1101 0.375 0.5 0.75

1.10011011 1.00111101 1.10011101

3 2

130 129

00111011001110110000000 00000000000000011101100

10000000 10000010

Your answer is correct.

The correct answer is:
To convert 19.6875 to IEEE 754 floating-point format step by step:

1. Sign bit: [0]
2. Convert to binary:
   2.1. Integer 19 → [10011]
   2.2. Fraction 0.6875 → 0.6875×2= [1] → [0.375]×2=[0] → [0.75]×2=[1] → [0.5]×2=[1] then the binary number is [1011]
   2.3. 19.6875 binary number is [10011].[1011]
3. Normalize: [1.00111011] × 2^[4]
4. Exponent: [4] + 127 = [131] → [10000011]
5. Mantissa: [00111011000000000000000]
6. IEEE 754 (32-bit): MSB [0][1][0][0][0][0][0][1][1][0][0][1][1][1][0][1][1][0][0][0][0][0][0][0][0][0][0][0][0][0][0][0] LSB

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Question 2

Partially correct

Mark 28.68 out of 30.00

To convert -0.1 to IEEE 754 floating-point format step by step:

1. Sign bit: → 1 ✓
2. Convert to binary (fraction only):
   0.1×2=0.2 → 0 ✓
   0.2×2=0.4 → 0 ✓
   0.4×2=0.8 → 0 ✓
   0.8×2=1.6 → 1 ✓
   0.6×2=1.2 → 1 ✓
   0.2×2=0.4 → 0 ✓ (repeats)

... → repeats 0001100110011... ✗

3. (In normalization, we need 1 on the left part of the decimal point, then we shift RIGHT until we get the 1.) Normalize:

1.1001100110011... ✓ × 2^ -4 ✓

4. Exponent: -4 ✓ + 127 = 123 ✓ → 01111011 ✓

5. Mantissa: 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✗ (rounded up to gain 23 bits)

6. IEEE 754 (32-bit): MSB 1 ✓ 0 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 0 ✓ 1 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✗ LSB

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1 0

0001100110011... 000110000000...

1.000110001100011... 1.100000000000000...

4

131

10000011

Your answer is partially correct.

You have correctly selected 65.

The correct answer is:
To convert -0.1 to IEEE 754 floating-point format step by step:

1. Sign bit: → [1]
2. Convert to binary (fraction only):
   0.1×2=0.2 → [0]
   0.2×2=0.4 → [0]
   0.4×2=0.8 → [0]
   0.8×2=1.6 → [1]
   0.6×2=1.2 → [1]
   0.2×2=0.4 → [0] (repeats)

... → repeats [0001100110011...]

3. (In normalization, we need 1 on the left part of the decimal point, then we shift RIGHT until we get the 1.) Normalize:
[1.1001100110011...] × 2^[-4]
4. Exponent: [-4] + 127 = [123] → [01111011]
5. Mantissa: [1][0][0][1][1][0][0][1][1][0][0][1][1][0][0][1][1][0][0][1][1][0][0][1][1][0][1] (rounded up to gain 23 bits)
6. IEEE 754 (32-bit): MSB [1][0][1][1][1][1][0][1][1][1][0][0][1][1][0][0][1][1][0][0][1][1][0][0][1][1][0][1] LSB

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